

AN ANALYSIS OF TOPOGRAPHIC INFLUENCES ON NON-SUPERCELL TORNADO
FORMATION ALONG THE CHEYENNE RIDGE, WYOMING-COLORADO BORDER

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ABSTRACT

Non-supercell tornadoes (NSTs), or landspouts, form through the stretching of pre-existing boundary-layer vorticity rather than mesocyclone-driven processes. Although typically weaker than supercell tornadoes, NSTs pose forecasting challenges due to their rapid formation, subtle radar signatures, and dependence on localized convergence zones. Along the eastern slope of the Rocky Mountains, terrain features such as the Palmer Divide and Cheyenne Ridge strongly influence low-level wind patterns, contributing to recurring mesoscale circulations such as the Denver Convergence Vorticity Zone (DCVZ). While NST development along the Palmer Divide is well documented, the role of the Cheyenne Ridge in shaping boundary-layer environments is largely under-researched. This study aims to investigate how the Cheyenne Ridge shapes surface convergence, vertical vorticity, and convective initiation associated with NST formation in southeastern Wyoming and northeastern Colorado. A long-term observational framework will integrate surface station data, Level II Doppler radar from KFTG and KCYS, ERA5 reanalysis fields, USGS elevation models, and historical tornado records from 1980–2024. Convergence and vorticity fields are derived from interpolated surface winds, terrain metrics are computed from digital elevation models, and NST frequency is analyzed using kernel density estimation and logistic regression. Expected results include identifying ridge-induced convergence zones, demonstrating statistically distinct NST environments, and mapping spatial hotspots of landspout occurrence. The research expands DCVZ-based understanding to a larger topographic setting and provides a new look into boundary interactions important for improving operational awareness and forecasting of NST hazards.

CHAPTER ONE: INTRODUCTION

Tornadoes represent one of the most hazardous and unpredictable severe weather phenomena in the United States, producing billions of dollars in damage and endangering lives each year (Brotzge & Erickson 2010; FEMA 2012). While most previous tornado research and forecasting practices focused on supercell thunderstorms, which generate the strongest and most destructive tornadoes, an additional class of tornadoes forms in environments without mesocyclones, deep convection, or strong synoptic forcing (Davies-Jones 1984; Bluestein 1985). These non-supercell tornadoes (NSTs), often referred to as landspouts, occur in more shallow convective environments and develop through the stretching of near-surface vertical vorticity along localized convergence zones. Although they are typically weaker than supercell tornadoes, NSTs are not a well-researched hazard due to their quick formation, subtle radar signatures, and frequent lack of warning.

NSTs are most likely to form in regions where terrain enhances boundary-layer circulations. The Denver metropolitan area, situated along the eastern slope of the Rocky Mountains, is one of the most well-documented examples of terrain-influenced severe weather. Interactions between southeasterly low-level (LL) flow, the Palmer Divide, and the Cheyenne Ridge produce mesoscale boundaries, including the Denver Convergence Vorticity Zone (DCVZ), that create vertical vorticity and support NST development (Szoke et al. 1984; Wilczak & Glendening 1988). Multiple damaging tornado outbreaks in the region, including those in 1981, 1988, and 2004, have demonstrated that local terrain can generate circulations even on days with weak instability and limited synoptic support (Crook et al. 1990; Szoke et al. 2006).

Despite many years of research on NSTs associated with the Palmer Divide and the DCVZ, there is little documented work that examines the broader terrain influences that may

support tornado development across the High Plains. The Cheyenne Ridge, a topographic feature extending along the Colorado-Wyoming border, shares characteristics with the Palmer Divide but has not been evaluated for its role in producing or enhancing NST-favorable environments. Sparse surface observation coverage, limited radar coverage, and low population density all contribute to a gap in the documentation and climatology of landspouts occurring north of the Denver area. As urban areas expand and as northeastern Colorado and southeastern Wyoming continue to experience convergent boundaries, the lack of understanding of the Cheyenne Ridge's meteorological influence presents both scientific and hazard-assessment challenges.

With the operational difficulty of detecting NSTs and the societal risk caused by NST events that form with little warning, particularly near urbanizing areas and infrastructure areas such as Denver International Airport (DEN), there is a need to better understand the terrain and boundary interactions that influence NST occurrence. Although previous studies made conceptual models of NST formation or simulated terrain-driven circulations, no published work utilizes long-term observational data, radar analyses, and terrain metrics to evaluate the Cheyenne Ridge's role within the broader DCVZ system.

The primary goal of this research is to examine how the Cheyenne Ridge modifies low-level convergence (LLC) and near surface vorticity and velocity fields that influence non-supercell tornado formation in southeastern Wyoming and northeastern Colorado. Being able to achieve the objective of understanding NST development will broaden the already established DCVZ research framework, improve understanding of terrain-influenced boundary processes, and strengthen the scientific basis for forecasting NST events in regions where shallow circulations recur. The study proposed herein will provide answers to the following research questions:

- How does the Cheyenne Ridge influence surface wind patterns, convergence, and vertical vorticity in the Denver Basin?
- Do NSTs forming near the Cheyenne Ridge form within environmental conditions, such as weak shear and modest instability, that are significantly different from supercell tornado environments?
- Are spatial patterns of NST frequency associated with topographic features such as slope or relative elevation along the ridge?

Understanding the mechanisms that contribute to landspout development is necessary for forecasting and hazard mitigation. NSTs form quickly beneath developing cumulus clouds, and they can cause localized wind damage, flight disruptions, and hazards to transportation and infrastructure, particularly within the growing urban and rural area east of Denver. As with other boundary-layer hazards, their impacts cannot be mitigated without an understanding of how surface processes, terrain features, and atmospheric conditions interact on the smaller scale. The research seeks to address these gaps by combining observational analyses, radar diagnostics, and climatological methods to evaluate the role of the Cheyenne Ridge in creating NST-favorable environments.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

Prior research on tornado genesis focused on supercell thunderstorms as the most prominent source of tornadoes. These tornadic storms, usually having a mesocyclone and a defined radar signature, represent the ideal tornado environment (Davies-Jones, 1984). Since the late 1970s, however, researchers began to recognize a different and distinct type of tornado that forms in environments that do not produce mesocyclones or strong synoptic forcing. These non-supercell tornadoes (NSTs), also known as landspouts (Bluestein, 1985), often occur beneath lower-level or shallow convection and in environments with weak vertical wind shear. The 1981 Denver outbreak supplied the first organized documentation of events that produced NSTs. With the use of the newly installed PROFS mesonet at that time, Szoke et al. (1984) found the Denver Convergence Vorticity Zone (DCVZ), which is a mesoscale boundary where southeasterly flow over the Palmer Divide generated the cyclonic vorticity needed to produce these storms. The tornadoes forming along this boundary raised questions to the in-place supercell structure and highlighted the importance of boundary layer interactions in tornadogenesis (Wilczak & Glendening, 1988; Crook et al., 1990).

Wakimoto and Wilson (1989) combined early observations of terrain-influenced storms into a conceptual model of NST development, while Brady and Szoke (1989) described their lifecycle from boundary establishment to vortex decomposition. More recent climatological work shows that these NSTs are not confined to Colorado, and that they occur across the High Plains (Kuster, Snyder, & Carr, 2015), Mexico (Monterde et al., 2023; León-Cruz et al., 2025), and even internationally (Brooks & Doswell, 2001); these findings confirm NSTs as a worldwide, though under-researched, hazard. Since NSTs can develop rapidly, often in

environments that are overlooked in forecasting for severe events, their detection and forecasting remain difficult (LaDue et al. 2012). Some occur without warning (Brotzge & Erickson, 2010), and their subtle radar signatures are not often detected beyond 50 km (McLaughlin et al., 2009). Being able to understand their mechanisms, terrain influences, and boundary layer processes is critical for both meteorology and the safety of the public.

2.2 Non-Supercell Tornadoes and Landspouts

Historically, tornado studies were mostly made up by the supercell process, where mesocyclones and organized storm structures were the necessary factors for tornadogenesis. Despite historical tornado studies, as Bluestein (1985) demonstrated during his observations in Oklahoma, tornadoes can also occur in some environments without supercells, often under much weaker synoptic and mesoscale forcing. Bluestein created the term "landspout" to describe these events, which allowed him to draw a direct comparison to waterspouts that form over warmer bodies of water. In both landspout cases, some vortices develop through the stretching of pre-existing vorticity near the surface, and not through the descent of rotation from aloft. Brady and Szoke (1988) were able to further research this understanding by showing evidence that landspouts were not isolated events, but instead a common type of tornado in northeastern Colorado. Their documentation of frequent, narrow, rope-like tornadoes that formed under lower or lacking instability was able to challenge the idea that all tornadoes require supercells to start tornadogenesis. A year later, Brady and Szoke (1989) confirmed the difference of NST storms in a case study showing that these NSTs developed in environments that had a closer resemblance to waterspout formation conditions than to traditional severe storm settings.

Being able to recognize NSTs as recurring rather than isolated events began a shift in tornado research. Wakimoto and Wilson (1989) examined decades of case reports and allocated

their field observations into a conceptual model of NST formation. Their study of 27 vortices across northeastern Colorado showed several factors that were common during all these NST occurrences. NSTs were usually associated with local boundaries, such as the DCVZ or outflow boundaries from other storms colliding, and they frequently formed beneath cumulus clouds before precipitation or lightning was present. Such factors separated them from supercell tornadoes, which usually occur under heavy precipitation and are often located in larger-scale severe weather outbreaks. Wakimoto and Wilson's (1989) work further enhanced the idea that NSTs were a separate category of tornadogenesis, which would require separate treatment and operational strategies.

2.2.1 Differences in Formation

The most noted difference between NSTs and supercell tornadoes is found in how they form. Standard supercell tornadoes form through the tilting of horizontal vorticity into the vertical within a rotating updraft. The mesocyclone structure of supercell tornadoes, which is a rotating updraft that can usually be seen on radar reflectivity, provides meteorologists with the ability to advance warning of potential tornado formation (Davies-Jones, 1984). Unlike supercell tornadoes, NSTs do not require a descending Rear Flank Downdraft (RFD) to provide the vertical stretching needed for tornadogenesis. Instead, NSTs form from the stretching of surface-based vertical vorticity (Figure 2.1), such as that associated with the DCVZ or a pre-existing outflow boundary. A tentative lifecycle model by Brady and Szoke (1989) described this process in four stages:

- 1: Boundary formation, which usually happens days or hours before tornadoes form.
2. Cumulus initiation directly over a localized vortex.
3. Stretching of the vortex into a tornado as the updraft deepens.

4. Dissipation as the parent storm matures or the boundary weakens.

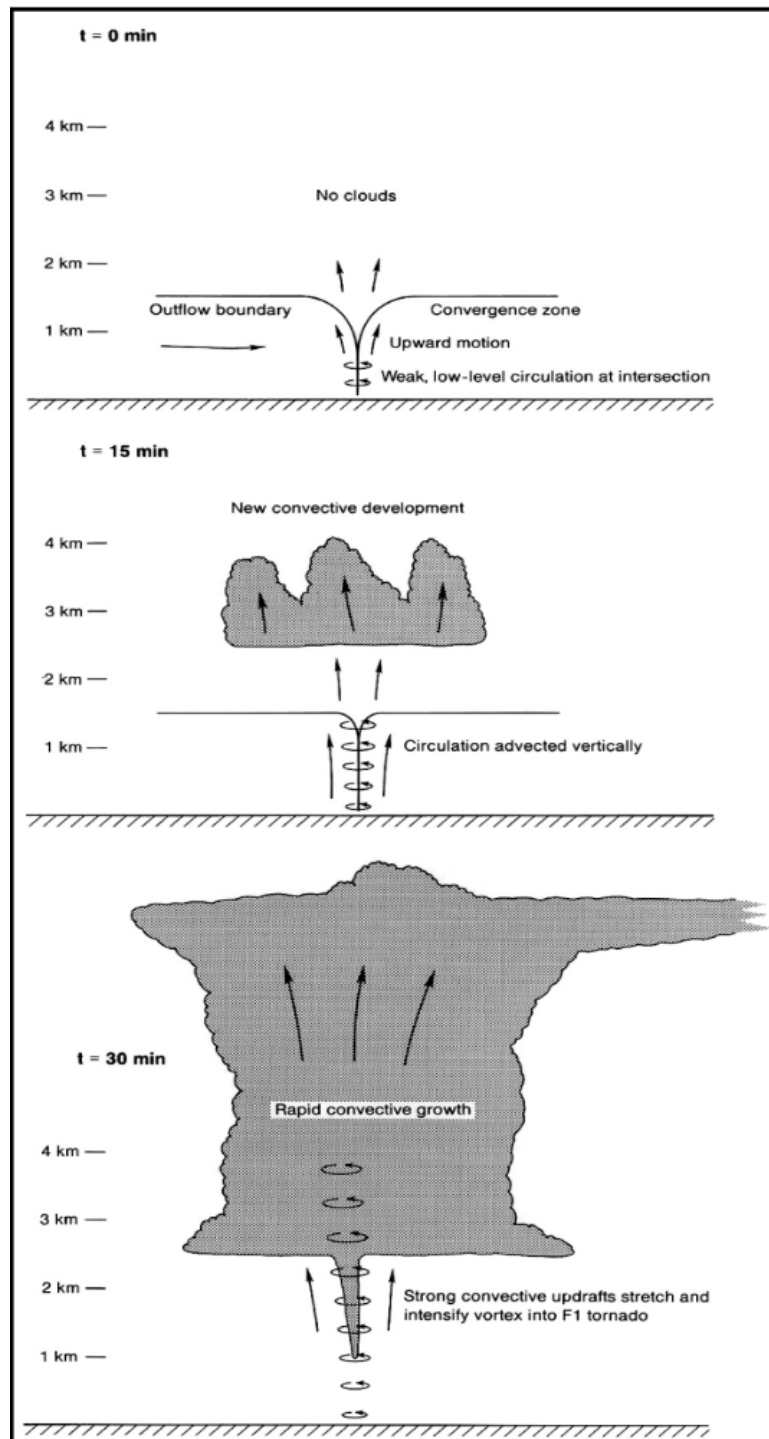


Figure 2.1: Schematic illustrating the life cycle of a nonsupercell tornado (Brady and Szoke 1989, p.3).

Roberts and Wilson (1995) offered support through Doppler radar scans of three NSTs. Their study showed that NSTs formed from strengthening boundary layer misocyclones (small, lower-level circulations), with little to no evidence of mesocyclone development, and this observation is also supported by numerical simulations, such as the Clark-Wilhelmson 3D storm model. Lee and Wilhelmson (1997a) used models to simulate these misocyclones that form along outflow boundaries, which shows that even the influence of weak boundaries could develop local circulations. In a similar study, Lee and Wilhelmson (1997b) looked at the life cycle of a family of tornadoes along a weak outflow boundary, and the results showed to be very similar to observed NST outbreaks where multiple landspouts occur chronologically. Modeling studies were able to expand the conceptual understanding of NSTs as a bottom-to-top process, which is different than the top-to-bottom processes that are central to supercell tornado development.

The question of whether NSTs should be considered different from supercell tornadoes or part of some sort of spectrum of tornadic processes is debated. Wilson (1986) argued that NSTs are representative of a separate class of tornadogenesis, one that is linked specifically to nonprecipitating lines of shear. Other studies, such as Brady and Szoke (1989), harped on their structural similarity to waterspouts, which would suggest that it may be better to view them as a terrestrial process rather than as variants of supercell tornadoes. The ongoing NST debate emphasizes the need to consider more than one pathway to tornadogenesis rather than assuming there is only a single, universal process.

2.2.2 Environmental and Atmospheric Conditions Favoring Landspouts

Analyses of environments consistently show that NSTs occur under weaker shear and modest instability, which are conditions that are traditionally considered unfavorable for tornadoes. Wakimoto and Wilson (1989) reported median Convective Available Potential Energy

(CAPE) values of 389 J kg^{-1} for their sample of 27 NSTs, which is well below the thresholds that are typically associated with supercells. In many cases, no lightning was detected, and the radar reflectivity was minimal, which means that these environments are not considered unstable by traditional severe weather indices. The 29 June 1994 Custer, South Dakota, landspout showed environmental traits with lacking instability values normally considered ideal for supercell tornado development. Hintz and Bunkers (1995) looked at this event, finding CAPE of 443 J kg^{-1} and a Bulk Richardson Number (BRN) of 5. Measured values of CAPE and BRN indicate weak instability and non-significant shear values, which are conditions that forecasters would not expect severe storm development. Despite this, the observed landspout produced winds estimated at EF1 intensity, overturning a 725 kg trailer and damaging a farmstead. Hintz and Bunkers' (1995) case reinforced that NSTs can, in fact, be dangerous, and that even in marginal environments, significant damage can occur. Analysis completed by Roberts and Wilson (1995) of the 15 June 1988 Denver outbreak provides another perspective, where intersecting outflows combined with the DCVZ to create multiple vortices, two of which produced EF2-level damage. Environmental CAPE was unimpressive, but the critical factor was the increase in horizontal shear at the intersection of the boundary. The 15 June 1988 Denver event showed that environmental instability by itself is insufficient to predict NSTs, but instead, local boundary interactions and surface vorticity are more dominant and important factors. Recent work has enhanced the understanding of NST environments apart from the Great Plains, and Monterde et al. (2023) analyzed an event with multiple NSTs in Mexico, where weak synoptic forcing but strong local convergence kicked off the formation of a family of landspouts. Monterde et al. (2023) showed that NSTs are not just occurring in the High Plains but instead are a broader occurrence in regions where topography or boundaries create elevated values of surface vorticity.

Similarly to Monterde et al., León-Cruz et al. (2025) examined tornadic environments in Mexico and found that NSTs often occur in monsoonal flow systems, which is again more of a result of surface convergence than of large-scale or synoptic forcing.

Kuster et al.(2015) documented a decade of weakly forced tornadoes in Oklahoma, and many of which fit the parameters to be considered an NST. Kuster et al. (2015) found that NSTs usually occurred on days with low shear and unimpressive CAPE values, which are environments that are often overlooked in forecasts. Brooks and Doswell (2001), in their global climatology of tornadoes, argued that NSTs account for a large fraction of tornadoes across the world, but particularly in regions where environments favorable for supercell development are rare. Together, these environmental analyses further show that NSTs are more likely to occur in conditions where supercell tornadoes are unlikely to form. NSTs' dependence on surface boundaries, stretching of vorticity, and frequent occurrence in low shear environments makes them a challenge in forecasting. Meteorologists cannot rely just on CAPE or shear values to identify NST potential, but instead, monitoring surface boundaries, local convergence, and diurnal heating is necessary.

2.3 Local Terrain Influences on Severe Weather

The occurrence of NSTs is heavily influenced by topography, which changes the shape of low-level wind patterns and creates the convergence boundaries that support tornadogenesis (Wilson 1986; Wakimoto and Wilson 1989). While studies of tornado environments look at synoptic scale forcing or large-scale instability, NST case studies focus rather on terrain influence (Szoke et al. 1984; Brady & Szoke 1989). The Palmer Divide, which is southeast of Denver, is one of the most widely studied terrain features that influences NST activity, but other ridges across the High Plains, including the Cheyenne Ridge along the Colorado-Wyoming

border and the Pine Ridge of Wyoming and Nebraska, show that these storms are not unique to Denver (Hintz & Bunkers 1995; Bann et al. 2003). By researching multiple ridges, researchers noticed a broader principle that terrain consistently creates localized convergence, enhances boundary vorticity, and increases the likelihood of landspouts.

2.3.1 Ridge-Influenced Storms

The Denver Cyclone, found in the lee side of the Palmer Divide, provides one of the clearest examples of how terrain influences NSTs. Before the 1980s, multiple outbreaks of small tornadoes in eastern Colorado confused forecasters because these events often happened in environments lacking the common factors of supercell thunderstorms. Zipser and Golden (1979) first hypothesized that a mesoscale circulation east of Denver might be able to explain these frequent tornadoes, suggesting that terrain-influenced flows were a key factor. An idea by Zipser and Golden (1979) was confirmed during the 3 June 1981 Denver tornado outbreak, when the newly deployed NOAA PROFS mesonet provided observational details. Szoke et al. (1984) documented the DCVZ, which was a boundary that formed when southeasterly surface winds hit the Palmer Divide and were turned northwestward, which created a cyclonic shear zone referenced in Figure 2.2. Multiple tornadoes formed along this boundary, which showed that local terrain influenced convergence could provide the necessary vertical vorticity for NST formation.

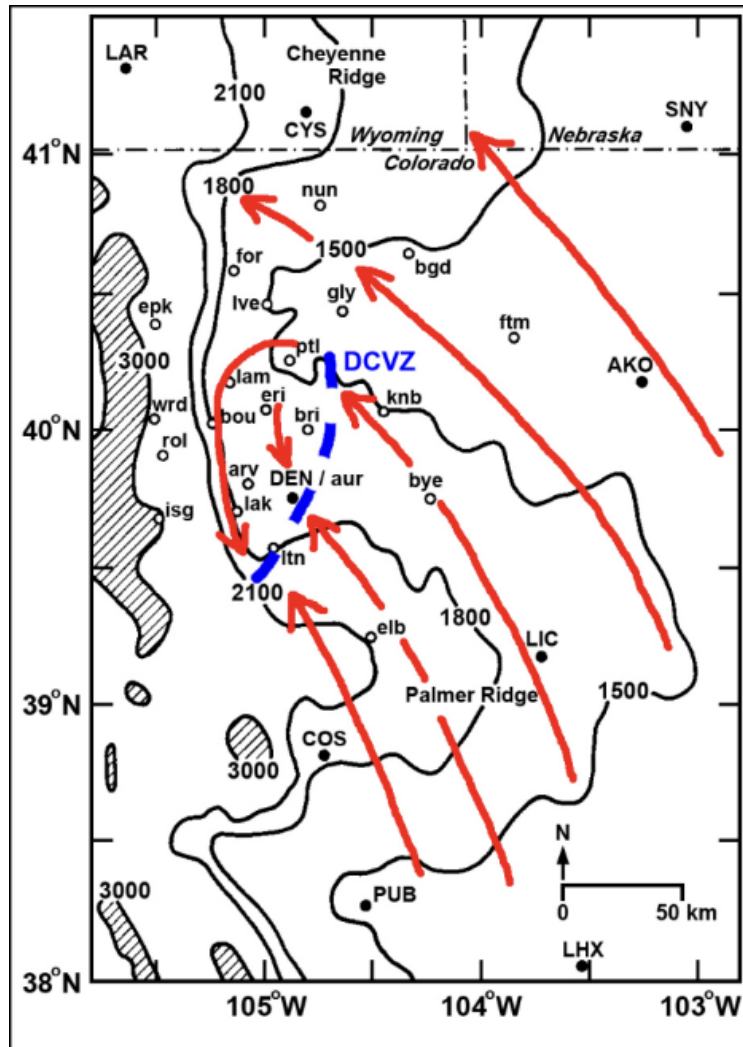


Figure 2.2: Schematic of the Denver Cyclone overlaid on a topographic map (m), with the PROFS mesonet stations shown by open circles and the standard NWS stations by darkened circles (Szoke et al. 2006, p. 3)

Later modeling work added to the case for the Denver Cyclone as a terrain influenced feature. Wilczak and Glendening (1988) used mixed layer modeling to show that southeasterly flow over the divide produced baroclinic zones, which created vorticity in the lee side, lining up with observations of the DCVZ. Crook et al. (1990) furthered the work of Wilczak and Glendening (1988) with three-dimensional numerical simulations, which showed that both stable

and unstable flows across the Palmer Divide could generate mesoscale vortices like the ones that were observed. Studies by Crook [et al.](#) (1990) showed that the presence of the ridge itself led the atmosphere to form cyclonic circulations that were not under the influence of large-scale instability or synoptic forcing.

Other case studies reinforced NST's connection with the Denver Cyclone. Wilczak and Christian (1990) studied a Denver Cyclone event and found that the mesoscale vortex closely matched the baroclinic theory. Their analysis showed that the convergence and cyclonic shear along the DCVZ were not momentary, but continuous, terrain-influenced features. Szoke and Augustine (1990) showed through climatology that the DCVZ showed up on nearly half of all summer days, with tornadoes forming on approximately 40% of those days during June. Without the influence of the Palmer Divide, the DCVZ would not exist. Szoke and Augustine's (1990) findings show an important point that terrain not only shapes tornado events but also creates environments that influence local tornado climatologies. In the city of Denver's case, the Palmer Divide explains why so many NSTs form along the Front Range.

2.3.2 Cheyenne Ridge Topography

North of the Palmer Divide, the Cheyenne Ridge is another example of terrain influencing severe weather events. Extending east to west along Colorado's border with Wyoming, the ridge rises above the plains around it, which redirects low level wind patterns and creates zones of local convergence. While the Palmer Divide receives more attention in meteorology, the Cheyenne Ridge contains many of the same features that make the Denver Cyclone possible. Bader (2008) looked at the Cheyenne Ridge from a geological perspective, connecting it to the Cheyenne Belt, which is a Proterozoic suture zone that shows the boundary between Archean and Proterozoic crust (Figure 2.3). A study by Bader (2008) described many

strike slips and uplift along this structural boundary, which produced a ridge that is both a geological and topographic feature. Despite Bader's work focusing on tectonics, there are many findings that can be applied to meteorology. These long-term structural boundaries shape topography, which in turn modifies airflow, especially near the surface. When southeasterly winds hit the ridge, they are moved upward and laterally, which creates conditions favorable for convergence and cyclonic vorticity. While the Cheyenne Ridge is not as heavily studied as the Palmer Divide, its geography suggests that it can produce a cyclone under the right conditions. Just as the Palmer Divide consistently influences the DCVZ, the Cheyenne Ridge may act as an influence for NST favorable boundaries.

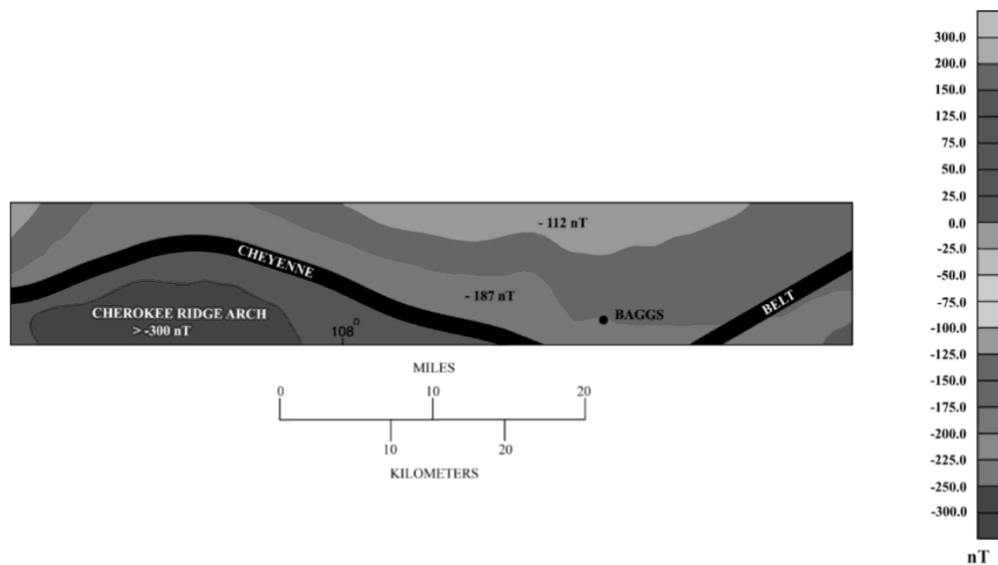


Figure 2.3: Aeromagnetic map of the Cherokee Ridge arch area showing magnetic contrasts along the Cheyenne Belt (Modified from Sims et al. 2001, as reproduced in Bader 2008, p. 6)

2.3.3 Landspout Favorability at the Cheyenne Ridge

The direct connection of landspouts along the Cheyenne Ridge is limited, mostly because of constraints in observations. Unlike the Denver area, which contains many available mesonets, Doppler radar coverage, and a large population to provide reports, the region north of Denver is not well instrumented and is less populated. Due to these lacking resources, tornadoes that occur are less likely to be observed, reported, or studied by researchers. Research focus bias may be part of the reason why the Palmer Divide is more studied in terrain-influenced NSTs. Despite less research, several lines of evidence support the ridge's influence on NST formation. Szoke and Augustine (1990) showed that the DCVZ produced tornadoes on almost half of the June days when it was formed. With the similar orientation and height of the Cheyenne Ridge, it is reasonable to assume that a similar environment to the Palmer Divide exists there. Bann et al.(2003) showed that the Pine Ridge of Wyoming and Nebraska was a focus for storm initiation because of the topography shaping winds into local convergence zones that produced severe storms. If the Pine Ridge can sustain storm initiation, the study aims to support the ideas that the Cheyenne Ridge could have the same terrain influence on NSTs.

Bader's (2008) analysis backs up this assumption. The ridge's history shows that it is a consistent feature that influences wind fields, and this consistency proves that there is a recurring role in how it shapes storms. Like how the Palmer Divide generates the DCVZ, the Cheyenne Ridge most likely produces local convergence that can be used to form convection to form NSTs. The lack of case studies says more about gaps in observation than about the ridge's irrelevance in storm development.

2.4 Boundary Layer Processes

Even though terrain features, such as the Palmer Divide or Cheyenne Ridge, provide influence for smaller-scale circulations, the processes that cause NST development happen within the atmospheric boundary layer. The boundary layer is where vertical vorticity is created, concentrated, and stretched into vortices for tornado development. Multiple studies show that three elements act together to determine whether or not a mesocyclone stays as a calm circulation, or whether it evolves into an NST. NST formation factors are surface vorticity, local convergence zones, and diurnal and seasonal timing. In contrast to supercells, which depend on convective dynamics and large-scale shear, NSTs are an effect of boundary layer interactions.

2.4.1 Surface Vorticity

Vertical vorticity at the surface is the most important process for NSTs. Wilson (1986) showed that vortices could form along nonprecipitation-induced shear lines, which are shallow boundaries not tied to thunderstorms. Wilson's finding was important because it showed that tornadoes do not require storm-scale circulations aloft to provide the necessary rotation. Instead, the boundary layer itself could create and supply the beginning vorticity values. Brady and Szoke (1989) emphasized the role of mesocyclones, which are defined as small circulations on the scale of 1-4 kilometers (km) in diameter. Mesocyclones are usually shallow, confined below ~1 km, and very difficult to see with operational radar. Despite this, mesocyclones provide the starting vortices that evolve into NSTs when stretched by an updraft. Roberts and Wilson (1995) supplied some of the best evidence of this process by using dual-Doppler radar to see the strengthening of boundary layer mesocyclones forming into tornadoes. Roberts and Wilson's data confirmed that tornadogenesis happens from bottom to top, which contrasts with the top-to-bottom idea that is usually associated with supercell tornadoes.

For NSTs, the stretching term is the most important in the vorticity equation. Szoke et al. (2006) saw vortices along the DCVZ with vertical vorticity of $\sim 0.01 \text{ s}^{-1}$, which is a sufficient value for tornadogenesis once there are cumulus towers formed overhead. These vertical velocity values are much smaller than ones found in a mesocyclonic tornado environment, but because NSTs operate on smaller scales, even small amounts of vorticity can be taken and quickly amplified. Lee and Wilhelmson (1997a) created a simulation that showed the initiation and life cycle of mesocyclones that formed along outflow boundaries, showing how they were stretched by convection. In another study, data showed the evolution of a family of tornadoes along a weak outflow boundary, which enhanced the tendency for multiple landspouts to form in a sequence in outbreak situations (Lee and Wilhelmson 1997b). Modeling explains events like the 4 October 2004 Denver outbreak, when ten NSTs formed in under an hour despite low values of CAPE ($\sim 900 \text{ J kg}^{-1}$). Once there is a favorable amount of vorticity, tornadoes can form until the boundary dissipates or weakens. Unlike supercells, NSTs are fundamentally reliant on surface vorticity. Supercells create their own storm rotation, while NSTs depend on already in place circulations near the surface in the boundary layer. The reliance on surface circulations makes them sensitive to small, local features that may not be recognized in larger scale analysis of storms.

2.4.2 Local Convergence Zones

Surface vorticity is rarely evenly distributed, and it is usually concentrated along local convergence zones. These zones can form in multiple ways, either through flow modified by terrain (e.g, the DCVZ), colliding outflow boundaries, differently heated surfaces, or the outflows of microbursts. Wilson and Schreiber (1986) showed that radar-observed convergence lines initiated convection and served as the starting point for NSTs. Wilson and Schreiber's

(1986) study found that these lines were widespread and often occurred before thunderstorm development in weakly forced environments. Roberts and Wilson's (1995) dual-Doppler analysis of the 15 June 1988 Denver outbreak provides a good example of this. On the day of their observation, two outflow boundaries collided with the DCVZ, which created multiple zones with high values of horizontal shear (Figure 2.4). Within an hour, four tornadoes developed, with two being rated at an EF2. The most important factor was not the amount of CAPE, but the local strengthening of shear and vorticity at the intersection of the outflow boundaries. Roberts and Wilson's case shows that boundaries, not just instability alone, are important in NST development.

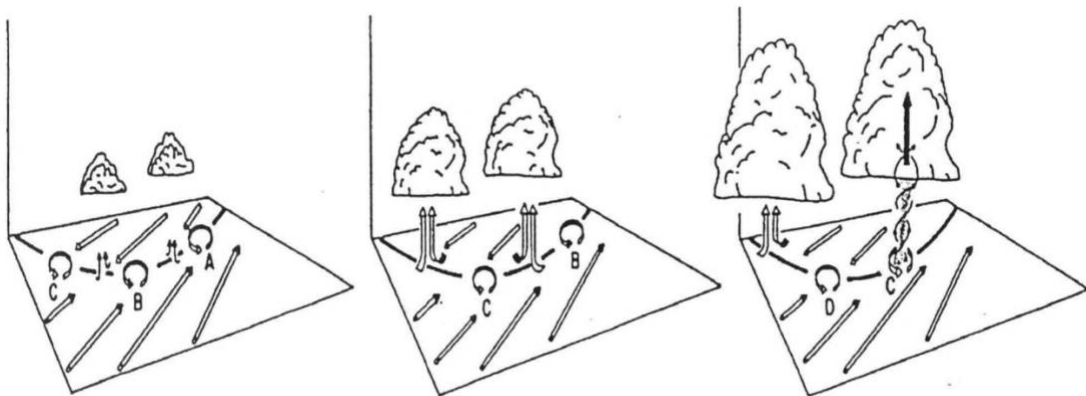


Figure 2.4: Schematic model of the development of a landspout showing convergence line, cyclonic shear, and misocyclones are labeled with letters A-D. (Wakimoto & Wilson 1989 / Hintz & Bunker's adaptation, p. 2)

Monterde et al. (2023) documented an outbreak with multiple landspouts in Mexico that happened under weak synoptic forcing, but strong local areas of convergence. Similarly, León-Cruz et al. (2025) showed that NSTs in Mexico were mostly tied to monsoonal flow areas, which concentrated vorticity along local convergence zones. These findings further show that

boundaries, not just synoptic forcing, provide the necessary factors for NST development across different countries. Other types of boundaries can also provide NST-favorable conditions. Atkins and Wakimoto (1991) examined wet microbursts in the southeastern United States and found that the outflow boundaries from these storms created lines of convergence with large amounts of boundary layer vorticity. Even though Atkins and Wakimoto's (1991) study studied downbursts, the application for NSTs is clear in that their outflows can create temporary convergence zones that can support vortices like landspouts. Atkins and Wakimoto's research broaden the environments where NSTs might form, other than the High Plains. Therefore, convergence zones provide a sufficient forcing mechanism to create an environment for mesocyclone development, and the ingredients required for initiation. Without convergence zones, surface vorticity may not be used, and with their influence, shallow convection can stretch these circulations into NSTs.

2.4.3 Diurnal and Seasonal Timing

NSTs show consistent patterns in timing, which shows their dependence on surface interactions and processes. Szoke and Augustine (1990) concluded that the DCVZ appeared on nearly half of summer days, but tornadogenesis was most common in June, when 40% of DCVZ days produced at least one tornado. Summer timing aligns with the maximum of surface heating and instability across the Front Range. From a diurnal standpoint, NSTs form in the afternoon and early evening, when boundary layer heating is at its maximum for the formation of cumulus clouds. The 29 June 1994 Custer, South Dakota landspout, analyzed by Hintz and Bunkers (1995), formed at 21:45 UTC, which is concurrent with the late afternoon preference that NSTs consistently show. Similarly, the 4 October 2004 Denver outbreak occurred in the afternoon hours, which again showed the predictability of diurnal timing, even when the environment and instability values were unimpressive. Kuster et al.(2015) studied a decade of weakly forced

tornadoes in Oklahoma and found that NSTs were unjustifiably common in late spring and early summer, with a more common maximum in the afternoon hours. Brooks and Doswell (2001) looked at this on a global scale and found that weak, landspout like tornadoes in Europe, Asia, and South America also formed in the afternoons during the warm season. León-Cruz et al. (2025) saw similar seasonal timing in Mexico, where NSTs often overlapped with monsoonal heating areas.

From an operational standpoint, these timing patterns cause unique problems, especially in forecasting and warning times. LaDue et al. (2012) stated that even though forecasters can anticipate the presence of boundaries that are favorable for NSTs during the day, the actual tornado formation often happens in narrow windows that are connected to cumulus initiation. Brotzge and Erickson (2010) concluded that many NSTs occur without National Weather Service warnings, mostly because their rapid formation and short life cycles are concurrent with the peaks of daytime heating. Brotzge and Erickson's (2010) findings tell that while diurnal and seasonal timing make NSTs predictable in a certain sense, they do not improve confidence in short-term warning processes.

2.5 Conclusion

Non-supercell tornado research shows that tornadogenesis can occur in environments very different from the classic supercell framework. Studies from Colorado, the Great Plains, Mexico, and other regions show that landspouts form through the stretching of surface-based vertical vorticity along boundaries influenced by terrain, differential heating, or interacting outflows. Environmental conditions consistently show weak deep-layer shear, modest instability, minimal radar reflectivity, and limited lightning activity during NST events, confirming the

bottom-up nature of their development. Terrain features such as the Palmer Divide, Cheyenne Ridge, and Pine Ridge modify low-level wind fields in ways that enhance convergence and vorticity, creating opportunities for boundary-influenced vortex formation. Boundary-layer processes, including surface vorticity concentration, convergence evolution, and diurnal heating, control the timing and intensity of NST formation and explain why NSTs develop quickly, show subtle radar signatures, and challenge warning operations.

Collectively, the reviewed literature shows a substantial gap in understanding how terrain features within the Front Range influence NST frequency and development. Research centered on the Palmer Divide forms a strong connection between ridge topography and cyclonic boundary formation, whereas the Cheyenne Ridge remains understudied despite similar topographic characteristics. A focused investigation of the Cheyenne Ridge's influence on surface vorticity, convergence structures, and convective initiation is therefore warranted. An investigation focused on the Cheyenne Ridge will clarify the role of terrain features in shaping NST environments and will support improved forecasting in regions where weakly forced convection frequently produces NSTs.

CHAPTER THREE: STUDY AREA

3.1 Regional Overview

Along the eastern base of the Rocky Mountains, the Denver metropolitan area represents a transition zone between the Front Range foothills and the High Plains of north-central Colorado. The city of Denver (39.7°N, 104.9°W) and the suburbs around it are within the Front Range Urban Corridor (FRUC), which represents a north to south band of urban development stretching from Pueblo, CO to Cheyenne, WY. The elevations in the FRUC range from approximately 1,500 meters (5,000 ft) on the plains to about 1,800 meters (6,000 ft) near the southern and western highlands (USGS, 2024). The elevation change within this urban area influences local circulations and zones of convergence.

The Denver-Aurora-Lakewood metropolitan region homes a population of over three million people (U.S. Census Bureau, 2023), and urban expansion from Denver continues eastward into the plains, where range and farmland are being turned into residential neighborhoods, distribution centers, and industrial parks. Many of these growing communities, such as Brighton, Commerce City, Watkins, and Bennett, are within or near the area where NSTs form along the Denver Convergence Vorticity Zone (DCVZ). Because NSTs usually develop with little to no warning and can form in populated or economically important areas such as Denver International Airport (DEN), the need to understand their environment is important in both the scientific and societal settings. The combination of local topography, urban growth, and convective activity makes the Denver region an optimal area for studying locally terrain-influenced severe weather and to improve forecast accuracy for NST events.

3.2 Topography and Terrain Features

The Denver area's terrain influences local wind patterns and storm development in the area, specifically within three regions. These three influential topographic features include the Front Range, the Palmer Divide, and the Cheyenne Ridge. All these topographic features form a shallow, bowl-shaped area known as the Denver Basin (Figure 3.1). To the west, the Front Range foothills rise to elevations more than 2,400 meters (7,874ft), which produces downslope wind accelerations and lee side convergence (Szoke et al. 2006). To the south, the Palmer Divide stretches east to west between Denver and Colorado Springs and rises several hundred meters above the surrounding plains. The Cheyenne Ridge separates the South Platte River Basin to the north from the Arkansas River Basin to the south, which acts as a barrier to the southerly flow in the region. When moist southeasterly winds hit the divide, they are lifted and redirected northward, which creates a zone of horizontal shear and low-level convergence that becomes the DCVZ (Wilczak & Christian 1990; Szoke et al. 2006).

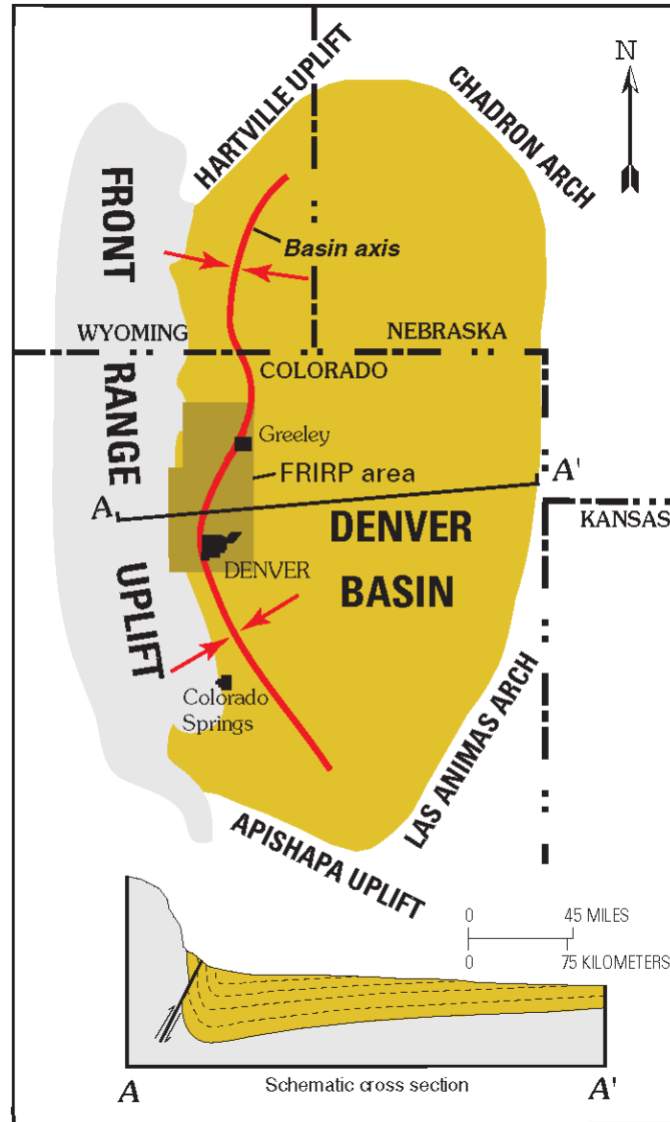


Figure 3.1: Location of the Denver Basin (Knepper 1996, USGS Circular 1219)

The Cheyenne Ridge, which is located along the Wyoming-Colorado border near Cheyenne, Wyoming, is another local topographic feature that influences local storm development (Bader, 2008). The Cheyenne Ridge blocks and redirects northeasterly and easterly flow, which helps to channel surface winds southward into the Denver Basin (Wilczak & Glendening 1988; Crook et al. 1990). During DCVZ events, air moving south along the

Cheyenne Ridge converges with air flowing north near the Palmer Divide, which produces a local cyclonic circulation east of Denver (Wilczak & Christian, 1990, Szoke et al., 2006;). Three terrain features, the Cheyenne Ridge, Palmer Divide, and Front Range, provide local topographic influence for one of the most well-documented mesoscale circulations in the United States (Wakimoto & Wilson 1989; Szoke & Augustine, 1990) (Figure 3.2).

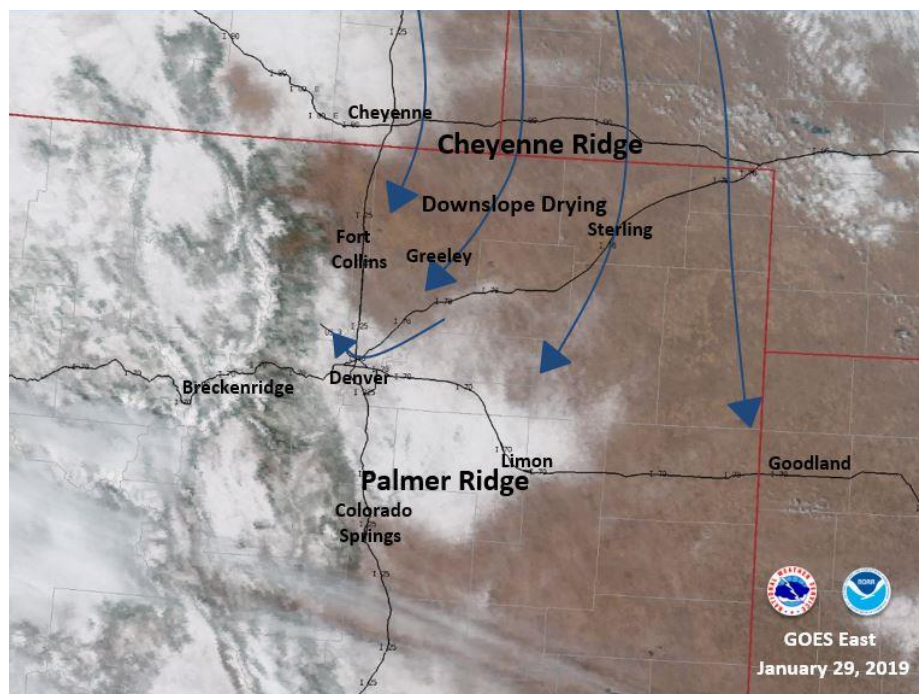


Figure 3.2: Cheyenne Ridge influencing downslope drying in the Denver area (NWS Boulder 2019).

3.3 Meteorological Significance

The region near Denver has a semi-arid climate, which means there are large daily temperature ranges and moderate precipitation averaging about 380 mm annually. Most convection happens between the months of May and August, which lines up with the highest climatological influence of severe weather along the Front Range. During this period of May-

August, the southerly to southeasterly surface winds commonly bring low-level moisture westward from the Great Plains. Even with weak synoptic conditions, the southerly to southeasterly surface winds will still interact with the local terrain to produce local areas of enhanced convergence and vorticity. Approximately one-third of summer days produce a circulation within the DCVZ, and around 40 percent of DCVZ days in June report associated tornado reports (Szoke and Augustine, 1990). Contrastingly to classic Great Plains supercell tornadoes, most are NSTs, which are short-lived, narrow vortices that form when local convection utilizes and stretches local areas of vertical vorticity along a surface boundary. Although NSTs are usually weak, these tornadoes can cause localized damage to structures, vehicles, and power lines, and cause hazards to both aviation and transportation. The region's urbanizing plains and proximity to DEN cause the potential impacts of these events to be more emphasized, and it further enhances the need to better understand the mechanisms that drive these NST events.

3.4 The Denver Convergence Vorticity Zone (DCVZ)

The DCVZ is a local mesoscale boundary that forms when southerly or southeasterly low-level wind interact with the terrain of the Denver Basin (Wilczak & Glendening 1988; Crook et al. 1990). First observed in the early 1980s through the NOAA/PROFS mesonet, the DCVZ is a narrow band of surface wind convergence and local cyclonic vorticity, which typically extends from DEN northward toward Greeley, CO (Szoke et al. 1984; Szoke and Augustine 1990). The localized circulation is usually visible in radar and satellite data as a small vortex, which is better known as the Denver Cyclone (Wilczak and Glendening 1988). The DCVZ forms due to vorticity generation and stretching. As local southeasterly winds cross the

Palmer Divide, local baroclinic vorticity is created by increasing temperature and pressure gradients. Horizontal vorticity in the zone of convergence is then tilted vertically by convective updrafts and amplified by convergence near Denver (Wakimoto and Wilson 1989). The Cheyenne Ridge also contributes to local northeasterly inflow, which closes the circulation and sustains a cyclonic flow that has the potential to circulate for several hours (Crook et al. 1990).

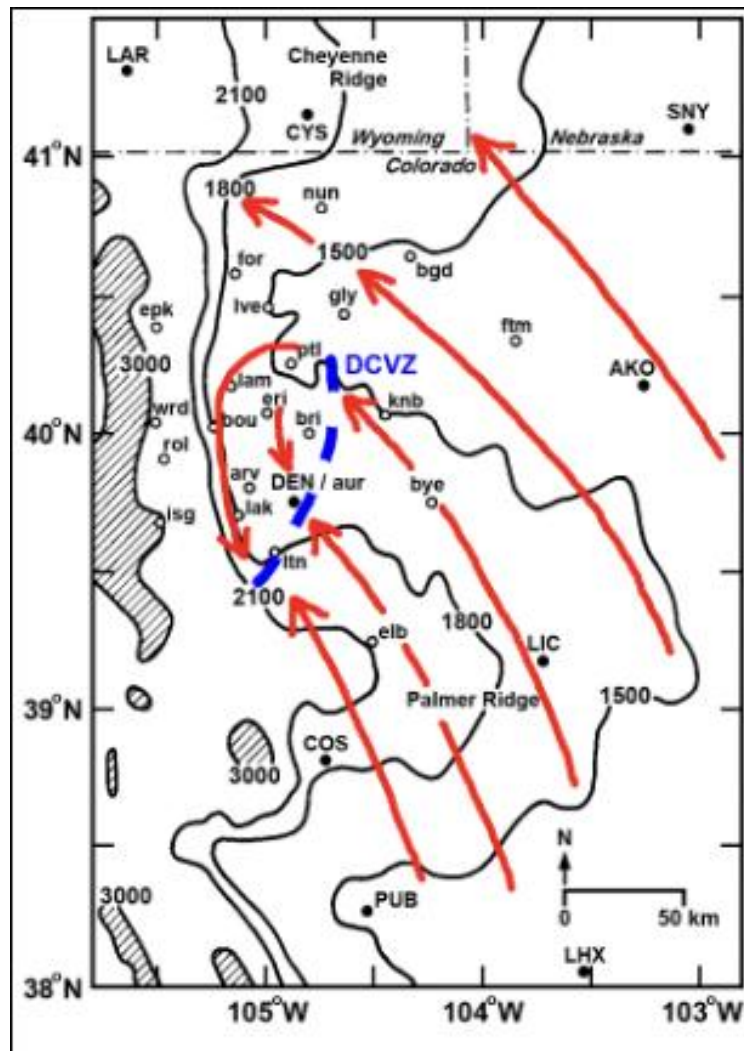


Figure 3.3: Schematic of the Denver Cyclone overlaid on a topographic map (m), with the PROFS mesonet stations shown by open circles and the standard NWS stations by darkened circles (Szoke et al. 2006, p. 3).

The DCVZ is known for producing NST outbreaks, including the June 3, 1981, Denver event, which first led to the identification of the DCVZ (Szoke et al. 1984). Additional notable events influenced by the DCVZ include the June 15, 1988, metro tornadoes, several of which reached F2 intensity (Roberts & Wilson 1995; Wilczak & Christian 1990), and the October 4, 2004 (Szoke et al. 2006), Denver International Airport outbreak, which produced ten tornadoes within an hour. Case studies associated with historic NST events show that even without the influence of synoptic forcing, local terrain and boundary interactions can produce hazardous weather (Wakimoto and Wilson 1989; Wilczak & Glendening 1988). The DCVZ represents an example of local terrain-influenced mesoscale dynamics that directly affect an urban region. Understanding the DCVZ's structure and the way it varies provides insights into NST development, forecasting challenges associated with them, and the larger relationship between topography, boundary layer processes, and convective hazards along the Front Range of Colorado.

CHAPTER FOUR: METHODOLOGY

The study proposed herein will utilize a correlational and observational design to examine how mesoscale and local topographic processes associated with the Cheyenne Ridge influence the frequency and formation of NSTs in southeastern Wyoming and northeastern Colorado. Since neither terrain nor atmospheric processes of the study area can be controlled, an observational framework will allow for the effective analysis of boundary layer interactions. The methods mentioned will integrate records of climatological data, surface observations, Doppler radar scans, reanalysis datasets, and elevation models to research relationships between terrain-induced convergence, low-level vorticity, and convective initiation (Wakimoto & Wilson, 1989; Szoke et al., 2006).

4.1 Data Sources

The study will utilize four primary data categories: surface observations, doppler radar scans, atmospheric reanalysis, and topographic data. Hourly surface data will be obtained from the National Centers for Environmental Information (NCEI) for all Automated Surface Observing System (ASOS) and mesonet stations located near the Cheyenne Ridge and east of it. Variables will include temperature, dewpoint, wind speed, wind direction, and pressure recorded at each station. Each dataset will be downloaded and checked for continuity and corrected for outlying or missing values. Wind direction will be converted to zonal (u) and meridional (v) components to allow for the vector analysis of convergence and vorticity (Wakimoto & Wilson, 1989).

Level II Doppler radar data will be collected from the Front Range/Denver (KFTG) and Cheyenne (KCYS) WSR-88D radar sites through the NOAA Weather and Climate Toolkit. Base

reflectivity and base velocity products from the 0.5° tilt scan will be re-fitted to a 1 km horizontal resolution using the Python Atmospheric Radiation Measurement Radar Toolkit (Py-ART) (Helmus & Collis, 2016). Only radar scans with beam heights below 1.5 km above ground level (AGL) will be used to make certain that boundary-layer circulations are measured accurately

ERA5 reanalysis data from the European Center for Medium-Range Weather Forecasts (ECMWF) will provide synoptic scale atmospheric context. The dataset has a horizontal resolution of 0.25° and an hourly temporal resolution (Hersbach et al., 2020). Variables extracted will include temperature, humidity, and wind at standard pressure levels. Derived parameters will include convective available potential energy (CAPE), convective inhibition (CIN), and lifting condensation level (LCL), as defined by Bolton (1980), along with bulk wind shear at 0–1 km and 0–6 km, following Doswell and Brooks (1993). Topographic data will be acquired from the U.S. Geological Survey (USGS) National Map at 30 m resolution. The digital elevation model (DEM) will be resampled to 500m to match the meteorological grid. Slope, aspect, and relative elevation will be calculated to assess how variations in terrain influence convergence near the surface and vortex generation (Bader, 2008).

Historical tornado and NST event data from 1980 to 2024 will be collected from the NOAA Storm Events Database (NCEI, 2024). The events will be verified using radar imagery, surface observations, and National Weather Service documentation (ie., storm reports). Tornadoes associated with mesocyclones will be classified as supercell events, while near boundary vortices without mesocyclones will be considered as an NST (Brady & Szoke, 1989; Szoke et al., 2006). Some cases will be excluded from statistical analysis to maintain dataset integrity, such as cases of landspout development while under the influence of a mesocyclone.

4.2 Boundary Diagnostics and Processing

All the collected datasets will be standardized to an hourly temporal resolution. Radar scans occurring within about 5 minutes of surface observation times will be connected to ensure temporal consistency. Surface winds will be interpolated to line up with radar scan times (Szoke et al., 2006). Surface winds will be interpolated spatially to a 0.1° grid using inverse distance, a method that estimates values at unknown locations by using samples at known value points. Surface winds will be interpolated by inverse distance weighting with a 50 km search radius and a minimum of four contributing stations. The resulting wind fields will be used to compute horizontal convergence and vertical vorticity values through centered difference calculations. Convergence will be defined as:

$$C = -\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right)$$

Equation 1: Convergence

and vorticity as:

$$\zeta = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}$$

Equation 2: Vorticity

Time periods with convergence values greater than or equal to $5 \times 10^{-5} \text{s}^{-1}$ will be designated as boundary-active, which follow the limits established for Plains mesoscale boundaries (Wakimoto & Wilson, 1989). Radar reflectivity and velocity data will be analyzed to

identify shallow convergence lines and shear zones associated with terrain influenced boundaries. Velocity gradients exceeding 12 m s^{-1} across distances of less than 3 km will be considered localized low-level (LL) shear features (Brady & Szoke, 1989). Radar data beyond 70 km from each site will be excluded to avoid beam overshooting, which can limit detection of shallow circulations as tilts increase angles from the surface (Wilczak & Glendening, 1988).

ERA5 soundings will be used to calculate thermodynamic indices as well as LCLs and levels of free convection (LFCs). CAPE and CIN will be computed using the virtual temperature correction, and LCL will be determined using Bolton's (1980) formulation. Bulk shear will be calculated as the difference between layer mean winds for the 0–1 km and 0–6 km layers. All the processed datasets will be stored in NetCDF format with complete metadata and coordinate system information to aid in reproducibility.

All analyses and visualizations will be conducted using Python (Jupyter Notebook) and ArcGIS. Figures will include landspout frequency maps, convergence frequency composites, and flow diagrams summarizing data. All processing scripts and datasets will be version-controlled using GitHub and documented with full metadata to support transparency and reproducibility (Helmus & Collis, 2016).

4.3 Landspout Climatology and Frequency

Long-term climatology of NST events will be built, and event coordinates will be plotted in ArcGIS Pro 3.6 over terrain and slope maps derived from the DEM. Kernel density estimation (KDE) will be used to identify areas of higher NST frequency and will be normalized by spatial area (Szoke et al., 2006). The relationship between slope of topographic features, mean LL

convergence, and NST occurrence will be analyzed using logistic regression. The model will be expressed as

$$P_{event} = \frac{1}{1 + e^{-(\beta_0 + \beta_1 S + \beta_2 C)}}$$

Equation 3: Logistic Regression for NST Development

Where P_{event} is the probability of a landspout occurring within a given grid cell. To compare the characteristics of the environment, mean values of CAPE, CIN, bulk shear, and LCLs will be calculated for both NST and supercell tornado cases. Differences in the calculated mean values will be evaluated using independent T-tests ($\alpha = 0.05$). These comparisons will help determine whether NSTs form in statistically distinct environments characterized by weaker shear and more shallow convection (Szoke et al., 2006). Two case studies will also be examined in detail, including the June 29, 1994, Cheyenne Ridge event and the October 4, 2004, Denver outbreak. Each case will be reconstructed at 10-minute intervals using surface, radar, and reanalysis data. Radar velocity and surface convergence fields will be analyzed to visualize boundary evolution and vortex formation mechanisms.

4.4 Assumptions

Horizontal homogeneity will be assumed over scales of approximately 10 km, and Coriolis effects will be neglected due to the short-lived nature of mesoscale circulations (Wakimoto & Wilson, 1989). ERA5 will be assumed to correctly represent large-scale environmental conditions, despite limited boundary layer resolution (Hersbach et al., 2020). Radar scans will be restricted to within 70 km of the radar due to beam overshoot (Wilczak &

Glendening, 1988). Regression will assume independence among these events, even though temporal clustering during multi-hour outbreaks may occur (Szoke et al., 2006).

4.5 Limitations

Surface station coverage across southeastern Wyoming may reduce the resolution of calculated convergence fields. Station coverage limitations will be handled through validation of interpolated wind fields against ERA5 surface winds and through sensitivity testing of interpolation parameters. Radar detection of weak vortices will be limited to within 70 km of radar sites, and analyses will be restricted to this range. Temporal mismatches in the datasets will be reduced through time alignment (about 5 minutes) and interpolation when necessary. Tornado reports with uncertain classification or location errors greater than 5 km will not be considered after radar verification to maintain the reliability of the data set.

CHAPTER FIVE: CONCLUSIONS AND EXPECTED RESULTS

Non-supercell tornadoes represent an important but frequently overlooked component of severe weather hazards in the United States (Wakimoto & Wilson 1989). Unlike supercell tornadoes, which have been studied thoroughly and are supported by observational and forecasting methods, NSTs form quickly, under shallow convection, and occur in environments that would otherwise be considered low risk (Bluestein 1985). NST's small radar signatures, short lifetimes, and dependence on localized, boundary-layer features make them very difficult to forecast and predict (Roberts & Wilson 1995). The forecasting challenge is increased in regions where terrain modifies low-level wind fields, creating convergence zones and vortices that may support tornadogenesis even on weak-shear, weak-instability days (Wilczak & Glendening 1988). The Denver Basin, shaped by the Palmer Divide, Cheyenne Ridge, and Front Range (Figure 5), is an example of how terrain influences NST potential and why better scientific understanding is needed as populations expand into more vulnerable areas (Szoke & Augustine 1990).

The research described herein intends to understand how the Cheyenne Ridge influences the formation and frequency of NSTs by changing LL convergence and vertical vorticity along the Colorado-Wyoming border. Through the combination of surface observations, Doppler radar analyses, reanalysis datasets, terrain metrics, and past tornado reports, the study aims to determine whether the ridge acts as a recurring driver of NST-favorable environments like the DCVZ. The expected results of this work include the identification of terrain-influenced convergence zones, characterization of boundary-layer vorticity sources, and a better understanding of how NST environments are different from those associated with supercells. The

researcher anticipates that the NST climatology developed in this study will reveal “hotspots” along the ridge and show statistically significant associations between terrain features and NST occurrence.

This research will be one of the first to formally examine the Cheyenne Ridge’s meteorological influence on NST formation by using long-term observational and statistical methods. The results are expected to contribute to a better understanding of terrain-influenced tornadogenesis and to provide a foundation for improving forecasting in regions where shallow circulations occur but are easily overlooked. Ultimately, this work aims to enhance both scientific knowledge and operational awareness of NST hazards, particularly within the rapidly growing urban and rural areas of northeastern Colorado and southeastern Wyoming.

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